

Math 421

Monday, September 22

Announcements

- Homework 3 due Friday
- Grace's Drop-In Hours
 - Wednesday: 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4

Chapter 1: Introduction

Section 4: The Completeness Axiom

$$M_0 \in S, m_0 \in S$$

Maximum & Minimum

$$M_0 \in S, M_0 \geq s, \text{ for all } s \in S.$$

Def. Let S be a nonempty subset of $\mathbb{R}$. If S contains a largest element, M_0 , then we call M_0 the **maximum of S** and write $M_0 = \max S$.

Def. Let S be a nonempty subset of $\mathbb{R}$. If S contains a smallest element, m_0 , then we call m_0 the minimum of S and write $m_0 = \min S$.

Examples -

$$\bullet \{1, 2, 3\} \quad \max S = 3 \\ \min S = 1$$

$$\bullet [a, b] \quad \max S = b \\ \min S \text{ DNE.}$$

$$\bullet (1, 3] \quad \max S = 3 \\ \min S \text{ DNE} \\ \text{? does not exist.}$$

- $(1, 3] \subseteq (1, \infty) \subseteq [1, \infty) \subseteq [0, \infty)$
- $[1, 3] \subseteq [1, \infty) \subseteq [0, \infty)$

Upper & Lower Bounds

Def. Let S be a non-empty subset of $\mathbb{R}$. If a real number M satisfies $s \leq M$ for all $s \in S$, then M is an upper bound for S and the set S is **bounded above**.

- $\{1, 2, 3\} \quad M = 4, 5$
- $\{1, 2, 3\} \subseteq (-\infty, 4] \subseteq (-\infty, 5]$
- $S \subseteq (-\infty, M]$

Def. Let S be a non-empty subset of $\mathbb{R}$. If a real number m satisfies $s \geq m$ for all $s \in S$, then m is a lower bound for S and the set S is bounded below.

- $(1, 3] \quad m = 1, 0$

- $S \subseteq [m, \infty)$

★ m, M do not need to be elements of S .

Bounded

Def. Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be **bounded** if it is bounded above and bounded below.

$$\hookrightarrow S \subseteq (-\infty, M]$$

$$\hookrightarrow S \subseteq [m, \infty)$$

$$S \subseteq (-\infty, M] \cap [m, \infty) = [m, M] \quad \text{for } m, M \in \mathbb{R}$$

$$\bullet (1, 3] \subseteq [1, 3] \subseteq [-17, 451]$$

$$\bullet (1, 3] \subseteq [-5, 5]$$

Activity

& $\sup S$ and $\inf S$.

For each of the following sets find (if they exist) the maximum, minimum, an upper bound, and a lower bound

1. $\mathbb{Z}$
 $\times$ $\times$
 $\times$ $\times$

2. $\mathbb{Q}$
 $\times$ $\times$
 $\times$ $\times$

3. $\mathbb{N}$
 $\times$ $\times$
 $|$ $|$

4. $(a, b]$
 b b
 $\times$ a

5. $\{r \in \mathbb{Q} : 0 \leq r \leq \sqrt{2}\}$
 $\times$ $\sqrt{2}$
 0 0

6. $\{\frac{1}{n} : n \in \mathbb{N}\}$
 $|$ $|$
 $\times$ 0

Supremum & Infimum

$$M = \sup S$$

• M is an upper bound for S .
 $M \geq s$, for all $s \in S$.

• If M_1 is another upper bound for S ,

$M \leq M_1$
 - If $M_1 < M$, there exists $s_1 \in S$ such that $M_1 < s_1 \leq M$

Def. Let S be a nonempty subset of $\mathbb{R}$. If S is bounded above and S has a least upper bound, then we call it the **supremum of S** and denote it by $\sup S$.

Def. Let S be a nonempty subset of $\mathbb{R}$. If S is bounded below and S has a greatest lower bound, then we call it the infimum of S and denote it by $\inf S$.

• If $K < M$ there exists $s_1 \in S$ s.t. $K < s_1 \leq M$.
 $K \in \mathbb{R}$.

Group Activity/Discussion

❖ Give an example of the following:

- A bounded set that has a minimum value but no maximum value. *Anything of the form $[a, b)$ works.*

- A set that has no least upper bound (or its least upper bound is infinite).

$$\{2k : k \in \mathbb{N}\}$$

❖ When is it true that $\sup S \in S$ and $\inf S \in S$?

If S has a maximum value, M , then $M = \sup S$

If S has a minimum value, m , then $m = \inf S$.

❖ Are $\sup S$ and $\inf S$ unique?

Yes! Try using contradiction assume $s \neq t$ and $s = \sup S$, $t = \sup S$.